



WHAT IS CURSOR?

A cursor is a database object which is used to retrieve data from result set one row at a time. The cursor can be used when the data needs to be updated row by row.



Types of Cursors in SQL

1. Implicit Cursor
2. Explicit Cursor



Implicit Cursor

Whenever DML operations such as INSERT, UPDATE, and DELETE are processed in the database, implicit cursors are generated automatically and used by the framework. These types of cursors are used for internal processing and **can't be controlled** or referred from another code area.



Explicit Cursor

This type of cursor is generated whenever data is processed by a user through an SQL block. Generally, the use of the SELECT query triggers the creation of an explicit cursor and can hold more than one row but process just one at a time.

CURSOR LIFE CYCLE



- ▶ **Declaring Cursor**

A cursor is declared by defining the SQL statement.

- ▶ **Opening Cursor**

A cursor is opened for storing data retrieved from the result set.

- ▶ **Fetching Cursor**

When a cursor is opened, rows can be fetched from the cursor one by one or in a block to do data manipulation.

- ▶ **Closing Cursor**

The cursor should be closed explicitly after data manipulation.

- ▶ **Deallocating Cursor**

Cursors should be deallocated to delete cursor definition and released all the system resources associated with the cursor.

WHY USE CURSORS?



In relational databases, operations are made on a set of rows. For example, a SELECT statement returns a set of rows which is called a result set. Sometimes the application logic needs to work with a row at a time rather than the entire result set at once. This can be done using cursors.



What is @@FETCH_STATUS?

Returns the status of the last cursor FETCH statement issued against any cursor currently opened by the connection.

Return value	Description
0	The FETCH statement was successful.
-1	The FETCH statement failed or the row was beyond the result set.
-2	The row fetched is missing.
-9	The cursor is not performing a fetch operation.

CURSOR LIMITATIONS



- ❖ A cursor is a memory resident set of pointers -- meaning it occupies memory from your system that may be available for other processes.
- ❖ The cursors are slower because they update tables row by row.